Predictive Modeling for Course Demand and Revenue Forecasting on EduPro: A Data-Driven Framework for Strategic Course Planning

Nibhisha S. Kakrania
Master of Business Administration (MBA)
Business Analytics
Mumbai, India
Email: nibhisha@gmail.com

Abstract

The rapid growth of online learning platforms has generated unprecedented volumes of educational and transactional data. However, many organizations continue to rely on historical reporting rather than predictive intelligence when making strategic decisions regarding course development, pricing, instructor allocation, and revenue planning. This study develops a predictive analytics framework for EduPro, an online learning platform, to forecast course demand and revenue performance using course characteristics, instructor attributes, and transactional records.

The analysis integrates data from 3,000 learners, 60 instructors, 60 courses, and 10,000 transactions to identify factors influencing enrollment demand and revenue generation. Feature engineering techniques were applied to construct business-relevant indicators including price bands, instructor experience categories, revenue-per-enrollment metrics, and category-level performance measures. An interactive Streamlit dashboard was developed to support scenario analysis and strategic decision-making.

Results indicate that course pricing is the strongest driver of revenue generation, while course ratings demonstrate a positive relationship with enrollment demand. Artificial Intelligence, Business, and Project Management emerged as the highest-revenue categories, while Data Science, Finance, and Web Development generated the largest enrollment volumes. The proposed framework enables EduPro to transition from descriptive reporting toward proactive planning and evidence-based decision-making.

Keywords: Predictive Analytics; Revenue Forecasting; Educational Data Mining; Learning Analytics; Enrollment Prediction; Business Intelligence; Online Education; Streamlit Dashboard

SSRN eLibrary Subject Areas

• Information Systems & Economics

• Business Intelligence
• Data Analytics
• Educational Technology
• Learning Analytics
• Forecasting & Prediction

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1. Introduction

Digital learning platforms have transformed the global education landscape by providing scalable, flexible, and accessible learning opportunities. As competition within the e-learning industry intensifies, organizations must leverage data-driven strategies to optimize course offerings, improve learner engagement, and maximize revenue generation.

While traditional reporting systems provide valuable historical insights, they are inherently reactive. Educational institutions increasingly require predictive capabilities that allow them to anticipate future enrollment trends, identify high-performing courses, and allocate resources efficiently.

EduPro is an online learning platform offering courses across multiple categories and instructional levels. Despite maintaining extensive operational data, the organization lacks a structured forecasting framework capable of predicting future demand and revenue outcomes.

This research addresses this challenge by developing an analytics framework that integrates educational and transactional datasets to forecast course demand and revenue performance while identifying key business drivers.

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2. Problem Statement

EduPro currently relies primarily on historical performance metrics when making decisions related to:

- Course launches
- Pricing strategies
- Instructor onboarding
- Category expansion
- Resource allocation

Without predictive analytics capabilities, management faces uncertainty regarding future demand patterns and revenue potential.

The absence of forecasting models creates several operational challenges:

- Increased business risk during course launches
- Suboptimal pricing decisions
- Inefficient instructor allocation
- Limited visibility into future revenue opportunities
- Reduced ability to respond proactively to market changes

The objective of this study is to transform historical educational data into forward-looking intelligence that supports strategic planning.

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3. Research Objectives

Primary Objectives

1. Predict enrollment demand for individual courses.
2. Forecast course-level revenue generation.
3. Forecast category-level revenue performance.

Secondary Objectives

1. Identify key demand drivers.
2. Analyze instructor performance factors.
3. Support pricing optimization initiatives.
4. Develop an interactive decision-support dashboard.

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4. Literature Review

The growing adoption of online learning platforms has generated significant interest in Educational Data Mining (EDM), Learning Analytics (LA), and Predictive Analytics. These disciplines focus on extracting meaningful insights from educational datasets to improve learning outcomes, institutional performance, and strategic decision-making.

Romero and Ventura (2024) describe Educational Data Mining as a collection of computational techniques used to discover hidden patterns within educational environments. Educational data mining enables organizations to understand learner behavior, evaluate course effectiveness, and identify factors influencing educational success.

Learning Analytics extends this concept by emphasizing the measurement, collection, analysis, and reporting of data about learners and their learning environments. According to Baker and Inventado (2014), learning analytics can support decision-making by identifying trends, forecasting outcomes, and improving resource allocation.

Recent studies have demonstrated the effectiveness of predictive analytics in forecasting enrollment demand, student retention, academic performance, and institutional revenue. Chaka (2022) found that predictive models can significantly improve organizational planning by providing forward-looking intelligence rather than relying solely on historical reporting.

Within commercial online learning environments, predictive analytics has become increasingly important for pricing optimization, instructor allocation, and content planning. Educational organizations are increasingly using business intelligence tools and forecasting frameworks to improve operational efficiency and maximize profitability.

Despite significant advances in educational analytics, much of the existing literature focuses on student performance prediction and retention analysis. Comparatively fewer studies have examined the integration of course characteristics, instructor attributes, and transaction-level data for simultaneous enrollment and revenue forecasting.

This study contributes to the literature by developing a unified predictive analytics framework that combines educational data mining, business intelligence, and revenue forecasting techniques to support strategic decision-making within an online learning platform.

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5. Research Gap

Existing research in educational analytics primarily focuses on predicting student performance, academic achievement, learner retention, and course completion rates. While these studies provide valuable insights into learner outcomes, relatively limited attention has been given to the business and operational dimensions of online learning platforms.

Specifically, there is a lack of research examining the combined influence of course characteristics, instructor attributes, pricing strategies, and transaction-level behavior on both enrollment demand and revenue generation. Most existing studies evaluate educational effectiveness rather than organizational sustainability and commercial performance.

This study addresses this gap by developing an integrated analytical framework capable of forecasting course demand, estimating revenue outcomes, and identifying strategic drivers of platform performance. The resulting framework extends traditional educational analytics by incorporating business intelligence and decision-support capabilities relevant to modern online learning organizations.

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6. Dataset Description

The study utilized four integrated datasets obtained from EduPro's operational environment.

Users Dataset

- 3,000 learners
- Demographic attributes

- Enrollment participation records

Teachers Dataset

- 60 instructors
- Expertise area
- Years of experience
- Instructor ratings

Courses Dataset

- 60 courses
- Category
- Type
- Level
- Price
- Duration
- Course ratings

Transactions Dataset

- 10,000 transactions
- Revenue information
- Enrollment activity
- Payment details

Table 1. Dataset Summary

Dataset	Records
Users	3,000
Teachers	60
Courses	60
Transactions	10,000

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7. Methodology

Data Preparation

The analytical process began with merging course, instructor, and transaction data at the course level.

Data cleaning procedures included:

- Missing value treatment
- Duplicate inspection
- Data type validation
- Date standardization

Feature Engineering

The following derived variables were created:

Course Features

- Price Bands
- Duration Buckets
- Rating Categories

Instructor Features

- Experience Groups
- Rating Categories
- Expertise Match Indicators

Revenue Metrics

- Total Revenue
- Revenue per Enrollment
- Category Revenue

Demand Metrics

- Enrollment Count
- Category Enrollment Share

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8. Exploratory Data Analysis

Platform Revenue Performance

The platform generated total revenue of approximately:

Total Revenue

$911,323

from:

Total Transactions

10,000

This indicates strong commercial activity and provides a robust foundation for predictive modeling.

Revenue by Category

The highest-revenue categories were:

Category	Revenue
Artificial Intelligence	$202,751
Business	$181,528
Project Management	$169,103
Programming	$77,454
Data Science	$77,052

Artificial Intelligence generated the highest revenue contribution, accounting for approximately 22% of total platform revenue.

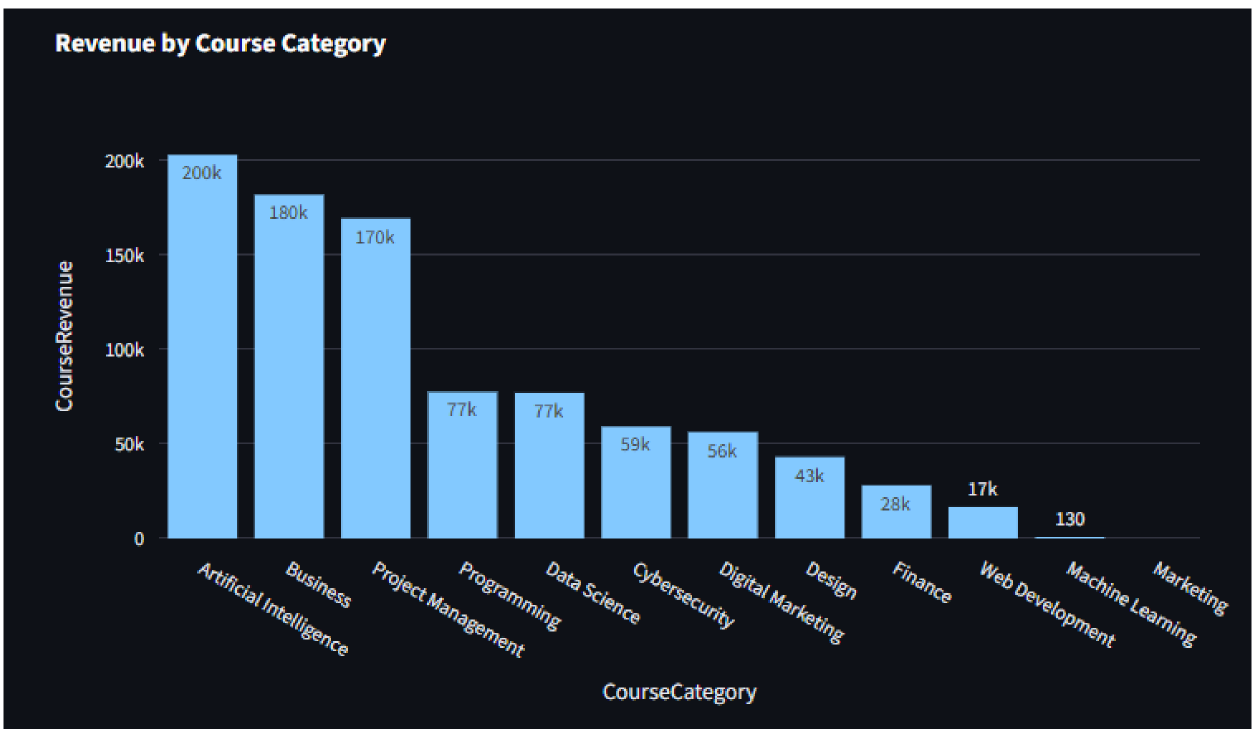

Figure 1. Revenue by Course Category

This visualization compares total revenue generated across all course categories available on the EduPro platform. The analysis identifies high-value categories contributing disproportionately to overall platform revenue, supporting resource allocation and investment decisions.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

Enrollment Demand by Category

The highest-demand categories were:

Category	Enrollments
Data Science	916
Finance	864
Web Development	844
Business	833
Project Management	829

This indicates that enrollment demand does not always align directly with revenue performance.

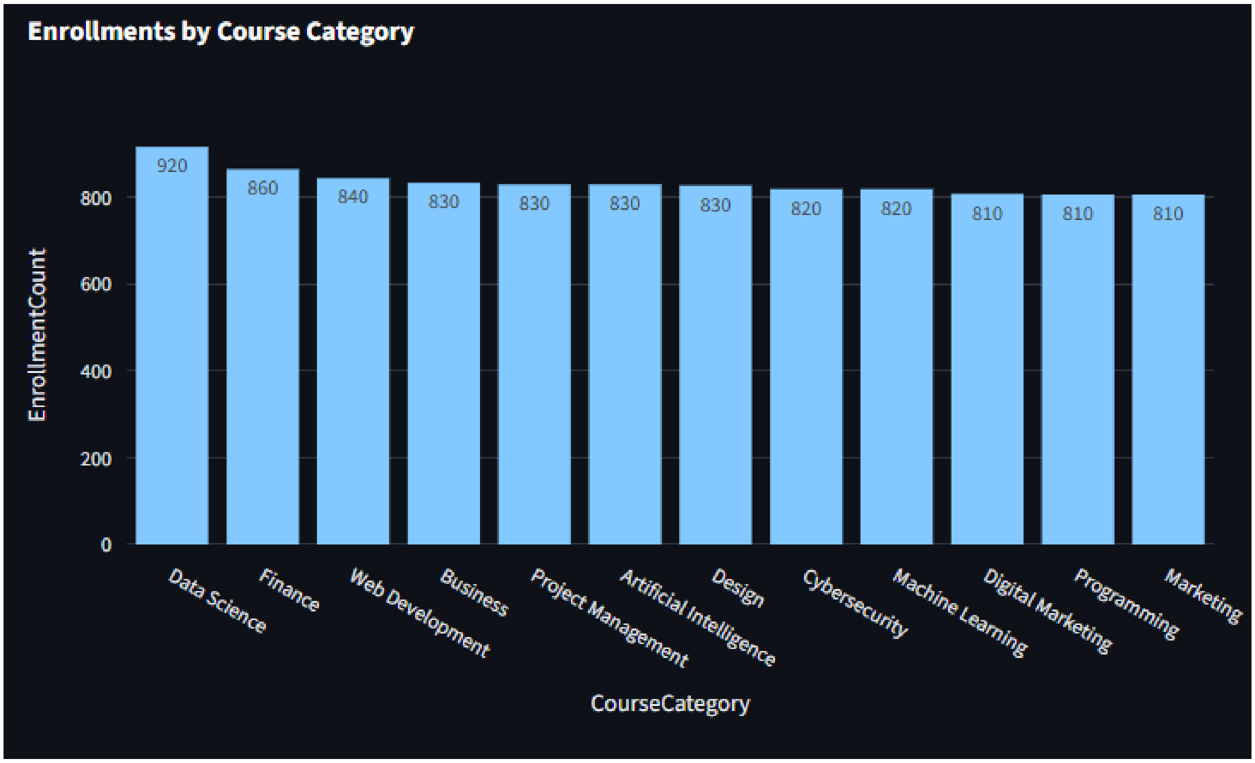

Figure 2. Enrollment Demand by Course Category

This figure displays enrollment volumes across course categories. Categories with strong learner demand may represent opportunities for content expansion, targeted marketing campaigns, and instructor recruitment.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

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9. Correlation Analysis

Correlation analysis was conducted to identify relationships between key variables.

Revenue Drivers

A very strong positive correlation was observed between:

Course Price and Revenue

$r = 0.997$

This suggests pricing is the dominant driver of revenue generation.

Demand Drivers

Course Ratings demonstrated a moderate positive relationship with enrollment demand:

r = 0.294

This finding suggests learners are more likely to enroll in highly rated courses.

Price Sensitivity

Course price showed a weak negative relationship with enrollment volume:

r = -0.163

Higher prices appear to reduce enrollment demand slightly while increasing overall revenue.

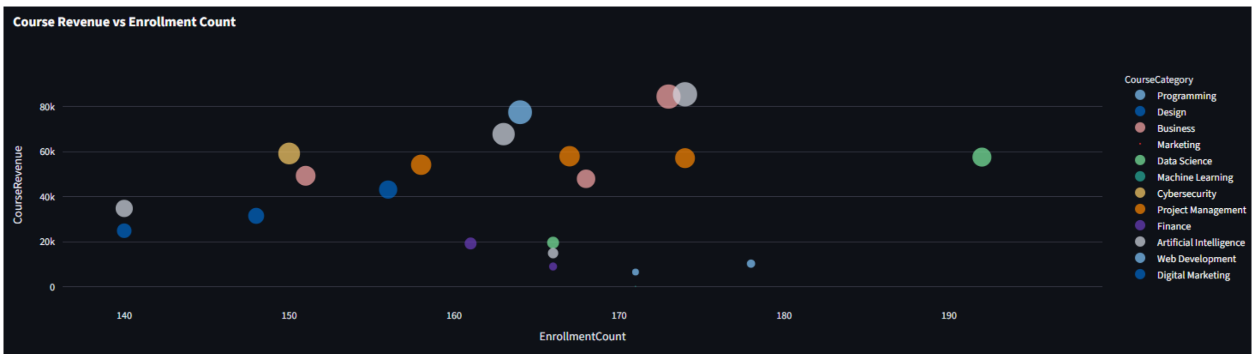

Figure 3. Course Revenue vs Enrollment Count

A scatter plot illustrating the relationship between enrollment demand and revenue generation at the course level. The visualization highlights courses that generate strong revenue despite moderate demand and courses with high demand but lower monetization efficiency.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

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10. Predictive Analytics Framework

The predictive framework consists of two primary forecasting components.

Enrollment Demand Forecasting

Demand predictions utilize:

- Course category

- Course level
- Course duration
- Course rating
- Instructor rating
- Instructor experience
- Expertise alignment

These factors collectively influence expected enrollment volumes.

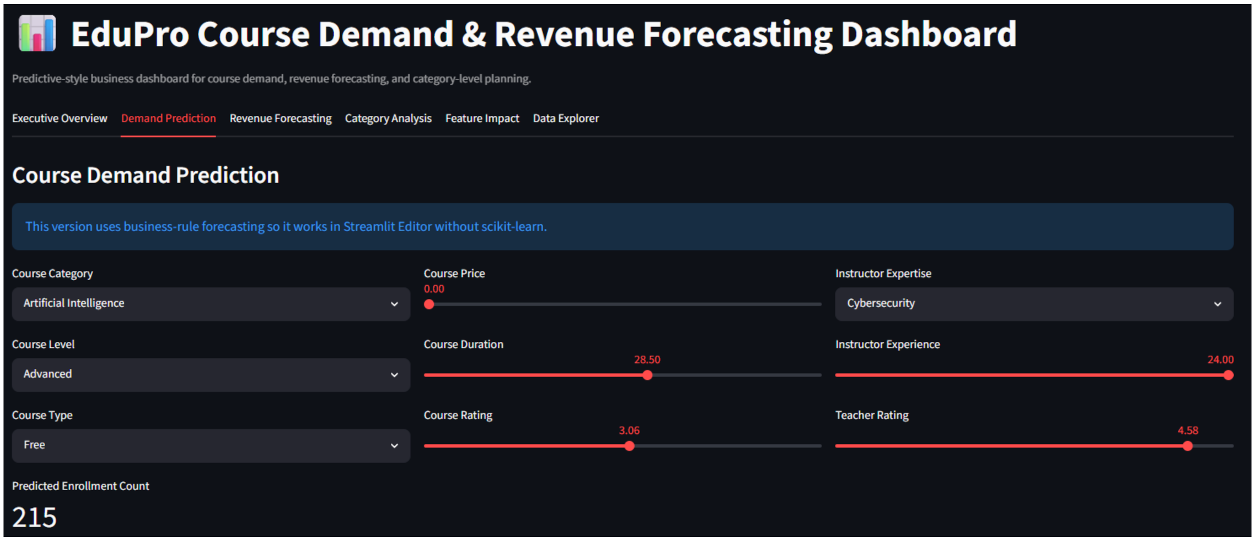

Figure 4. Course Demand Prediction Dashboard

The demand prediction interface allows users to modify course characteristics, including category, level, duration, instructor rating, and experience. The forecasting module provides projected enrollment estimates that support course launch planning and demand forecasting.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

Revenue Forecasting

Revenue predictions are derived from:

Revenue Forecast = Predicted Enrollments × Expected Revenue per Enrollment

This approach combines demand forecasting with pricing behavior and historical transaction performance.

Figure 5. Revenue Forecasting Dashboard

The revenue forecasting module estimates expected course revenue using pricing and instructional characteristics. This tool supports pricing optimization and financial planning by evaluating potential revenue outcomes under different scenarios.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

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11. Dashboard Development

An interactive Streamlit dashboard was developed to operationalize the analytical framework.

Dashboard Modules

Executive Overview

- Total Revenue
- Total Enrollments
- Average Revenue per Course
- Top Performing Categories

Demand Prediction

Interactive estimation of future enrollments based on selected course characteristics.

Revenue Forecasting

Revenue simulation using configurable pricing and instructor attributes.

Category Analysis

Comparative analysis of category-level demand and revenue performance.

Feature Impact Analysis

Identification of variables influencing enrollment and revenue outcomes.

Data Explorer

Interactive exploration of the processed analytical dataset.

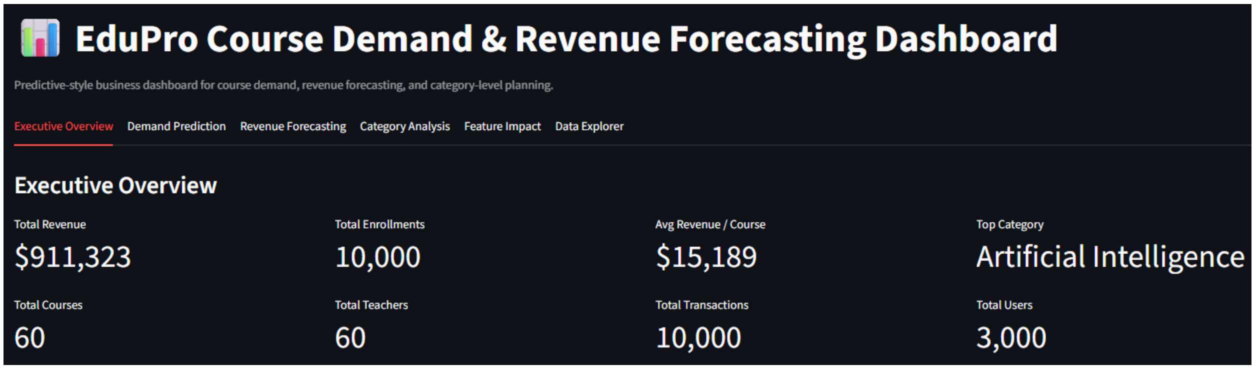

Figure 6. Executive Overview Dashboard

The Executive Overview dashboard provides a high-level summary of platform performance, including total revenue, total enrollments, average revenue per course, top-performing categories, and platform-wide activity metrics. The dashboard enables decision-makers to rapidly assess overall business performance and identify strategic growth opportunities.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

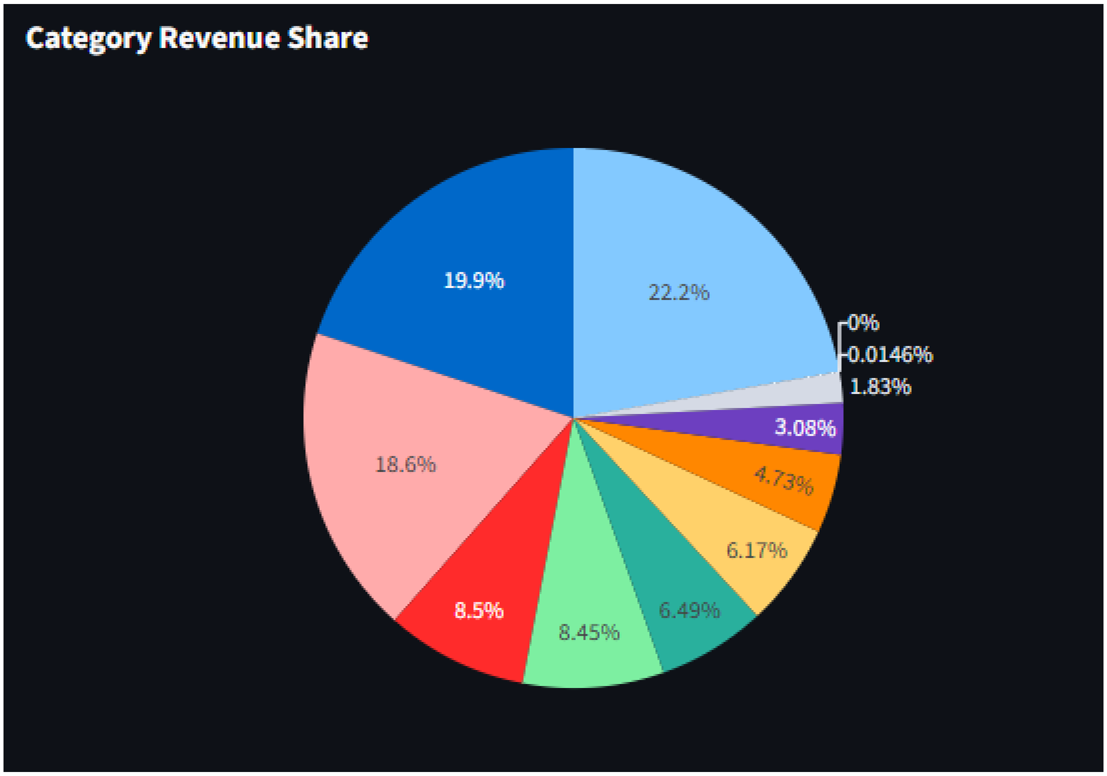

Figure 7. Category-Level Revenue Share Dashboard

A category-level visualization illustrating the proportion of platform revenue contributed by each course category. The figure highlights revenue concentration patterns and supports strategic portfolio management.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

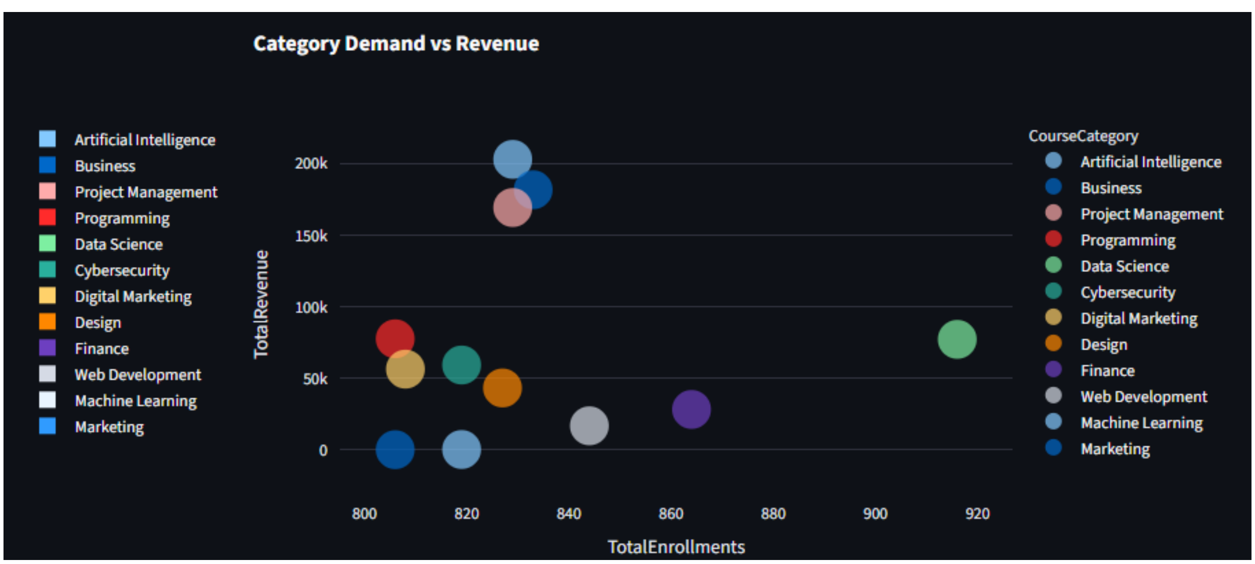

Figure 8. Category Demand vs Revenue Analysis

A comparative analysis of enrollment demand and revenue generation across categories. The visualization identifies categories that perform strongly across both demand and revenue dimensions.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

Table 2. Key Performance Indicators

KPI	Value
Total Revenue	$911,323
Total Transactions	10,000
Total Courses	60
Total Teachers	60
Total Users	3,000
Highest Revenue Category	Artificial Intelligence
Highest Demand Category	Data Science

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12. Key Findings

Finding 1

Artificial Intelligence courses generated the highest revenue contribution across the platform.

Finding 2

Data Science courses attracted the largest enrollment volumes.

Finding 3

Course pricing is the strongest predictor of revenue generation.

Finding 4

Course ratings positively influence enrollment demand.

Finding 5

Business and Project Management categories demonstrate strong performance across both demand and revenue dimensions.

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13. Managerial Implications

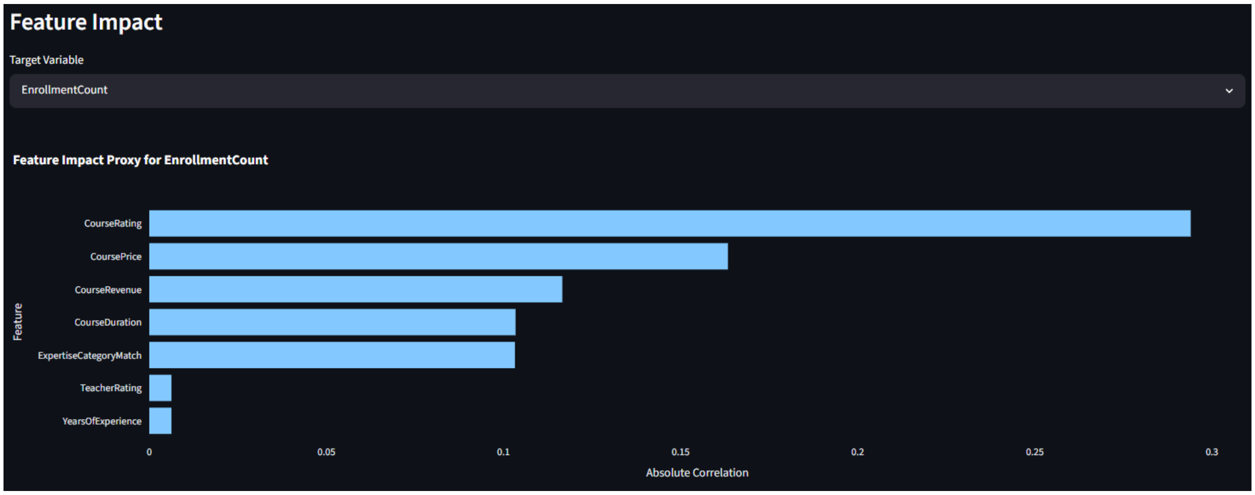

Figure 9. Feature Impact Analysis

This dashboard visualizes the relative influence of key variables on enrollment demand and revenue outcomes. Results indicate that pricing, ratings, instructor experience, and expertise alignment significantly influence business performance.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

The results provide several actionable recommendations.

Pricing Optimization

Management should implement data-driven pricing strategies to maximize revenue while minimizing enrollment decline.

Portfolio Expansion

Additional investment should be directed toward:

- Artificial Intelligence
- Business
- Project Management

due to their strong revenue performance.

Demand Growth

Marketing resources should target:

- Data Science

- Finance
- Web Development

to capitalize on strong enrollment demand.

Instructor Allocation

Highly rated instructors should be assigned to premium courses and strategic categories.

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14. Ethical Considerations

This study utilized anonymized educational and transactional data for analytical purposes. No personally identifiable information (PII) was collected, stored, or analyzed during the research process.

All analytical procedures were conducted in accordance with responsible data analytics practices and ethical research principles. The objective of the study was to improve organizational decision-making and operational efficiency rather than evaluate individual learners or instructors.

The findings presented in this paper are intended solely for educational, analytical, and strategic planning purposes.

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15. Limitations

Several limitations should be acknowledged.

1. Historical data may not capture future market changes.
2. External economic factors were not incorporated.
3. Competitor activity was unavailable.
4. Learner behavioral data was limited.

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16. Future Research

Future studies may incorporate:

- Random Forest Regression
- Gradient Boosting Models

- Deep Learning Forecasting
- Dynamic Pricing Optimization
- Student Retention Prediction
- Personalized Course Recommendation Systems

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17. Conclusion

This study demonstrates how predictive analytics can transform educational and transactional data into actionable business intelligence. By integrating course, instructor, and revenue data, the proposed framework enables EduPro to forecast demand, estimate revenue potential, and support evidence-based strategic planning.

The analysis identified significant relationships between pricing, ratings, enrollment behavior, and revenue generation. Artificial Intelligence emerged as the highest-revenue category, while Data Science generated the strongest enrollment demand. These findings provide valuable guidance for future course development, instructor allocation, and pricing decisions.

The resulting dashboard offers a scalable decision-support platform that enables EduPro to transition from reactive reporting to proactive planning, improving operational efficiency and long-term business performance.

The Streamlit dashboard developed as part of this project further enhances the practical value of the analytical framework by enabling interactive exploration of demand and revenue scenarios. By transforming complex analytical outputs into an accessible decision-support system, the dashboard bridges the gap between technical analysis and managerial decision-making. As online education continues to evolve, predictive analytics frameworks such as the one proposed in this study will become increasingly important for supporting sustainable growth, improving operational efficiency, and maintaining competitive advantage within the digital learning marketplace.

Appendix A

Data Explorer

	CourseID	CourseName	CourseCategory	CourseType	CourseLevel	CoursePrice	CourseDuration	CourseRating	EnrollmentCount	CourseRevenue	AvgTransactionAmount	UniqueLearners	TeacherID
0	CR00001	Python Basics	Programming	Paid	Beginner	472.28	11	4.74	164	77453.92	472.28	164	TC00042
1	CR00002	Java Programming	Programming	Free	Intermediate	0	37.7	2.43	149	0	0	149	TC00042
2	CR00003	C++ for Beginners	Programming	Free	Beginner	0	19.53	3.85	173	0	0	173	TC00040
3	CR00004	Advanced Python	Programming	Free	Beginner	0	45.13	2.88	154	0	0	154	TC00040
4	CR00005	Full Stack Development	Programming	Free	Beginner	0	28.68	1.28	166	0	0	166	TC00040
5	CR00006	Graphic Design Basics	Design	Paid	Advanced	276.37	48.84	2.72	156	43113.72	276.37	156	TC00042
6	CR00007	UI/UX Design	Design	Free	Advanced	0	48.68	2.61	178	0	0	178	TC00040
7	CR00008	Photoshop Mastery	Design	Free	Advanced	0	2.87	1.52	158	0	0	158	TC00042
8	CR00009	Web Design Fundamentals	Design	Free	Beginner	0	48.19	4.51	155	0	0	155	TC00042
9	CR00010	3D Modeling	Design	Free	Advanced	0	15.29	4.45	180	0	0	180	TC00040

Figure A1. Data Explorer Dashboard

The Data Explorer module provides access to the processed analytical dataset used throughout the study. The module enhances transparency and allows stakeholders to validate analytical findings independently.

Author's analysis using the EduPro dataset and Streamlit dashboard developed for this study.

Appendix B: Data Dictionary

Variable	**Description**
CourseID	Unique course identifier
CourseCategory	Course category classification
CourseType	Delivery format of the course
CourseLevel	Beginner, Intermediate, or Advanced
CoursePrice	Price charged for enrollment
CourseDuration	Course duration in hours
CourseRating	Average learner rating

TeacherID	Unique instructor identifier
TeacherRating	Instructor performance rating
YearsOfExperience	Teaching experience in years
TransactionID	Unique transaction identifier
TransactionDate	Enrollment transaction date
Amount	Revenue generated from transaction
EnrollmentCount	Number of enrollments per course
CourseRevenue	Total revenue generated by a course
RevenuePerEnrollment	Revenue efficiency metric

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